WHEN THE GROUPOID QUOTIENT OVER A FIELD IS THE SPECTRUM OF A FIELD

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; the quotient identification is delegated to an uncarded chapter lemma and declared external.

Allowed standard prerequisites (editorial).

- Presentation points. Properties of Spaces, lemma points-presentation (cross-chapter, named). Provides the two displayed points of U over k.
- Quotient sheaf of a groupoid over a field. lemma-quotient-groupoid-field (same chapter, NOT carded, declared external). Identifies X as the quotient sheaf.

Source reference. Original reference 04GG: focused-seeds/geometry/S9-decent-when-field/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 06RS: focused-seeds/geometry/S9-decent-when-field/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03BM: focused-seeds/geometry/S9-decent-when-field/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03JV: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 732–732. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 03K5: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2865–2865. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original theorem 086U: sources/stacks-decent-spaces/pinned-original-source.tex, source lines 2323–2323. Navigation pointer only: the counted

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proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space over S.*

(1) *If $|X|$ is a singleton then X is a scheme.*
(2) *If $|X|$ is a singleton and X is reduced, then $X \cong \mathrm{Spec}(k)$ for some field k.*

Proof. Assume $|X|$ is a singleton. It follows immediately from Theorem 086U that X is a scheme, but we can also argue directly as follows. Choose an affine scheme U and a surjective étale morphism $U \to X$. Set $R = U \times_X U$. Then U and R have finitely many points by Lemma 03JV (and the definition of a decent space). All of these points are closed in U and R by Lemma 03K5. It follows that U and R are affine schemes. We may shrink U to a singleton space. Then U is the spectrum of a henselian local ring, see Algebra, Lemma 06RS. The projections $R \to U$ are étale, hence finite étale because U is the spectrum of a 0-dimensional henselian local ring, see Algebra, Lemma 04GG. It follows that X is a scheme by Groupoids, Proposition 03BM.

Part (2) follows from (1) and the fact that a reduced singleton scheme is the spectrum of a field. □